

Understanding ML, DL, NLP
course

(1)

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Sec - 40

K means Clustering
Unsupervised ML

lec 1: K means Clustering
Geometric intuition

lec 2: - How to find
K values

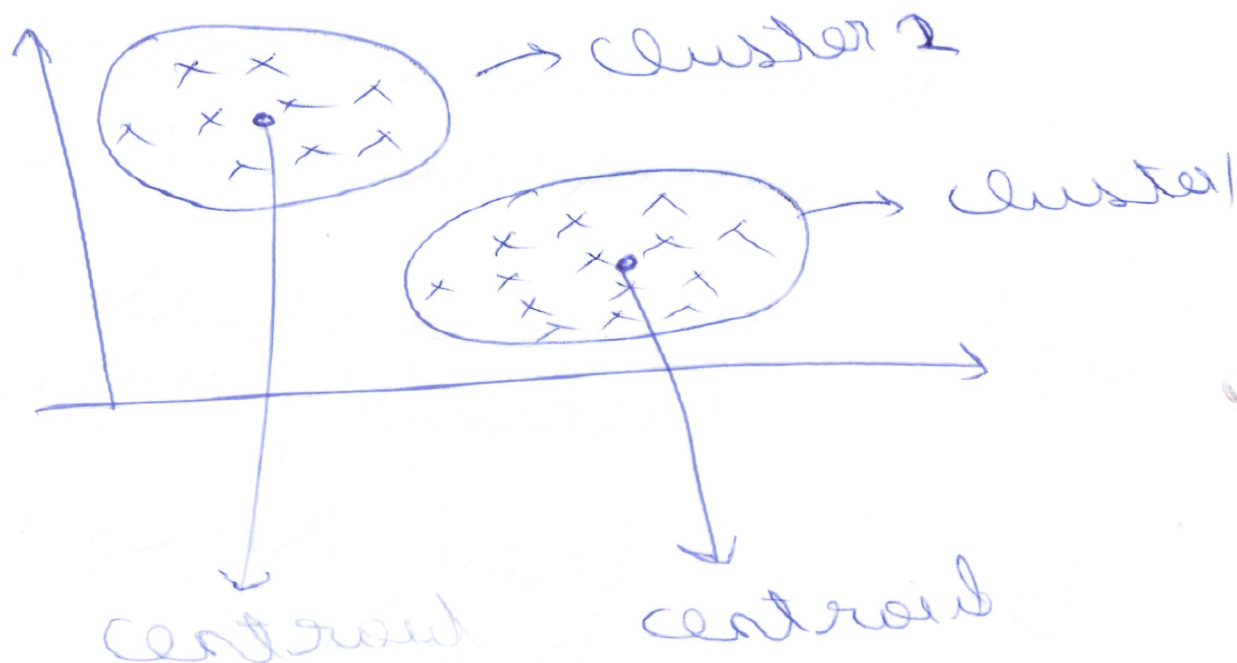
lec 3: Random initial
cluster centers



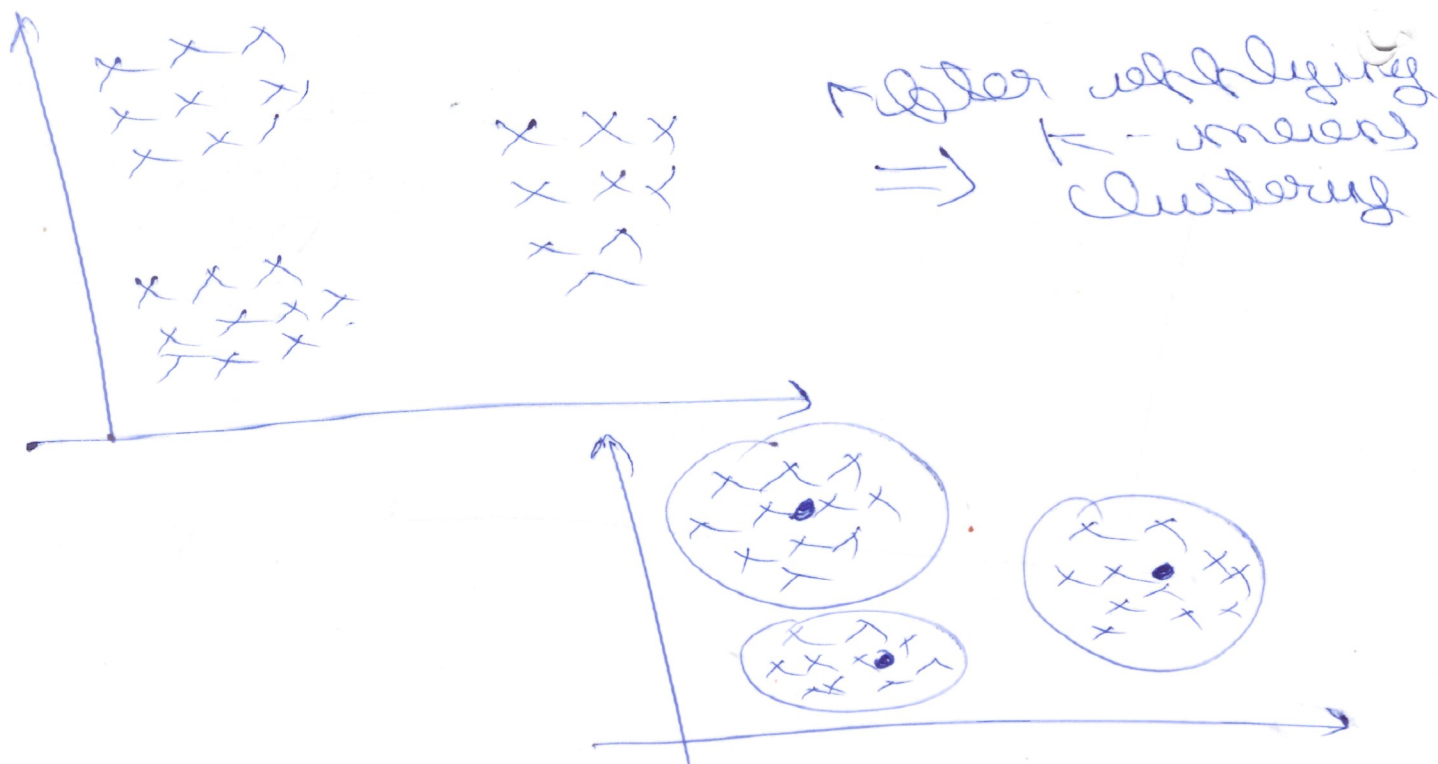
→ two
groups of
data
points

P.T.

after the application of
 means clustering we
 basically get two
 groups:



Now suppose if we look
 at another set of examples



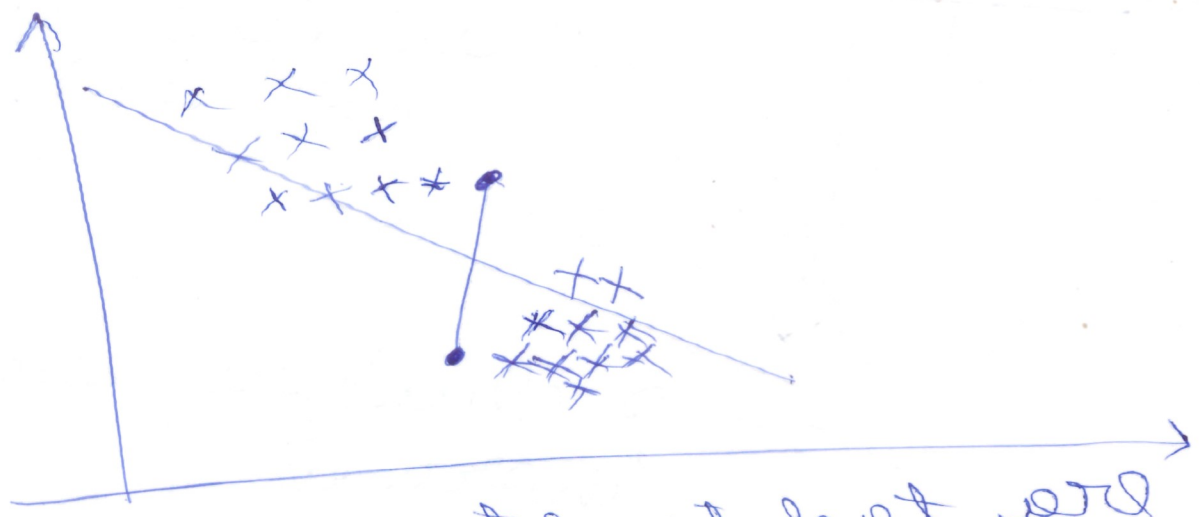
K means

① Initialize some ~~clusters~~ ~~centroids~~

Here let $K=2$

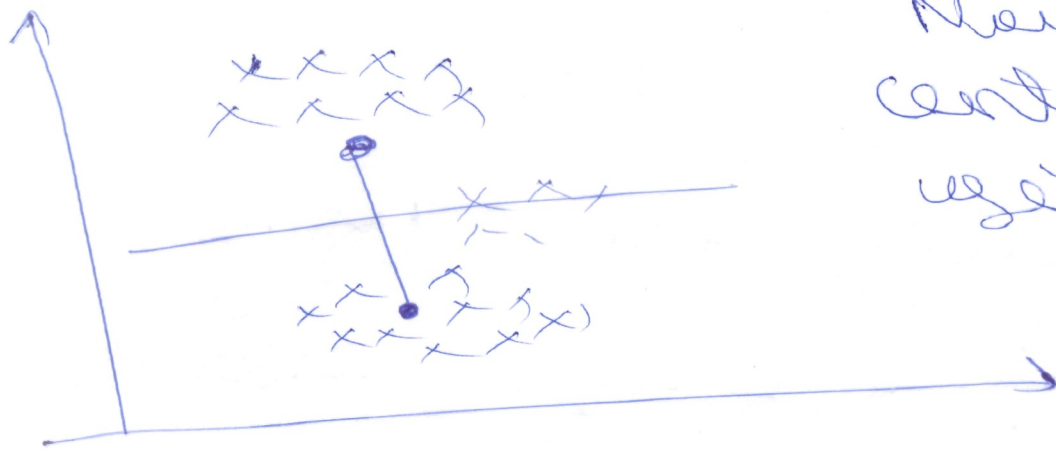
Centroids

② We find the distance of each point from the 2 centroids

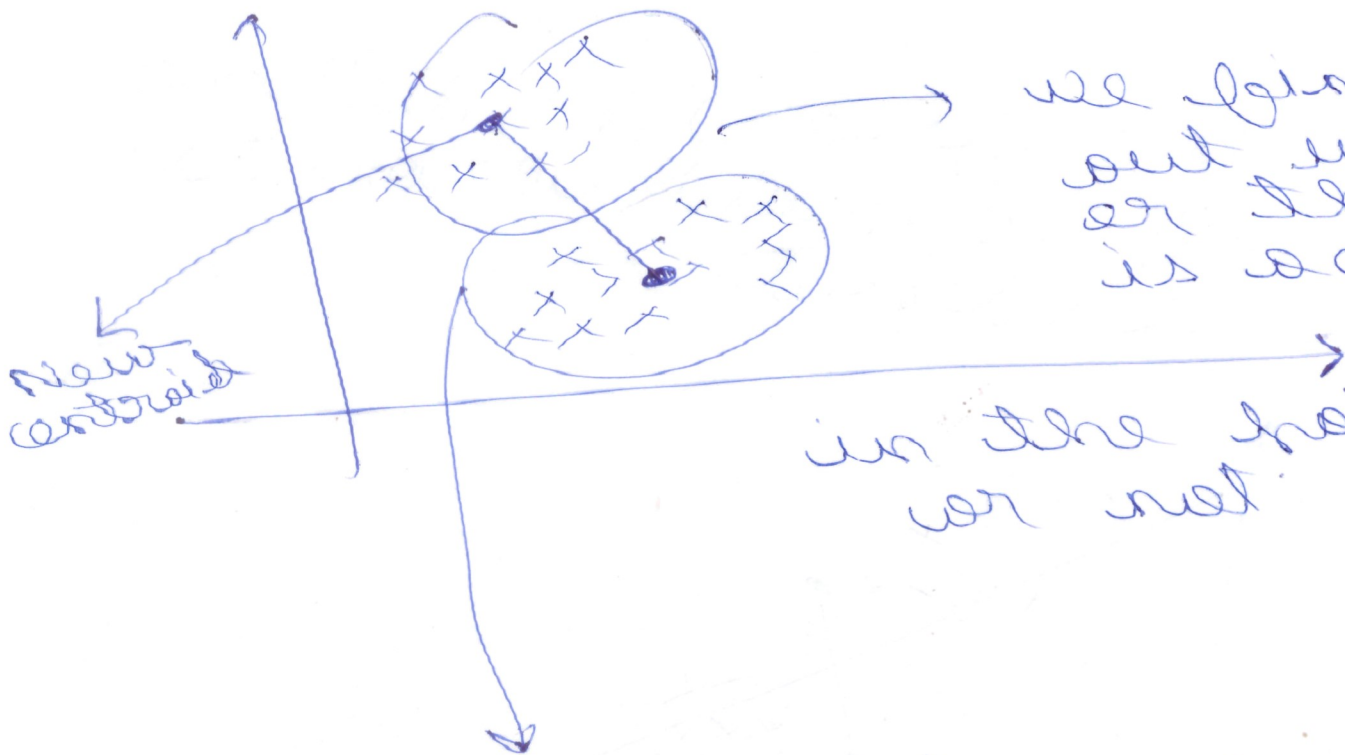


③ Points that are nearest to the centroids move

④ move the centroids
→ Average



Now, the
centroids
get updated



we find
out what
it is or
not

in the points
or not

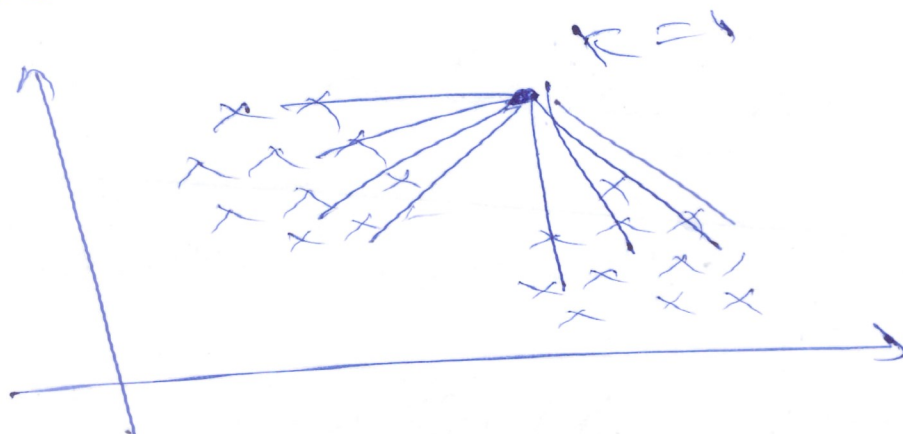
Here 2 clusters are
formed

How do we select the k values?

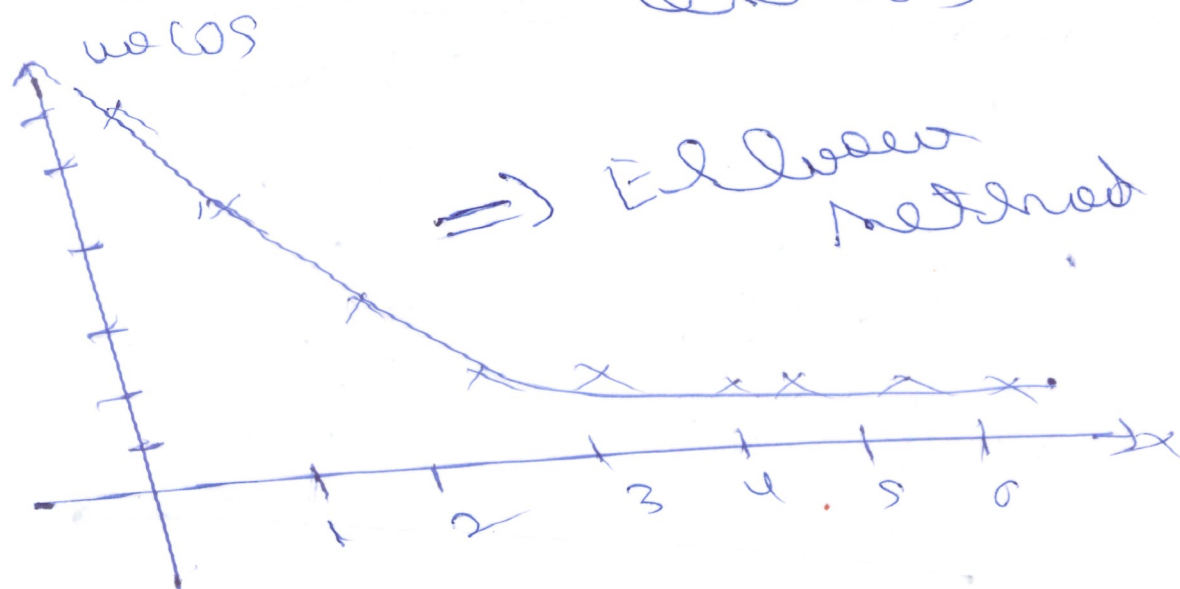
5

$wcss \Rightarrow$ within cluster, sum of squares.

Initial $k = 1$ to 20

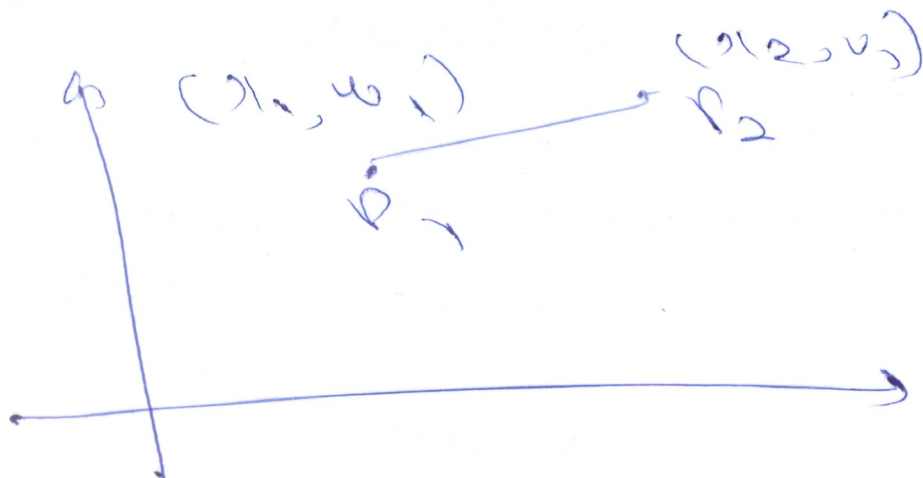


$$wcss = \sum_{i=1}^n (\text{distance of } i \text{th point to nearest centroid})$$



⇒ Euclidean Distance

(6)



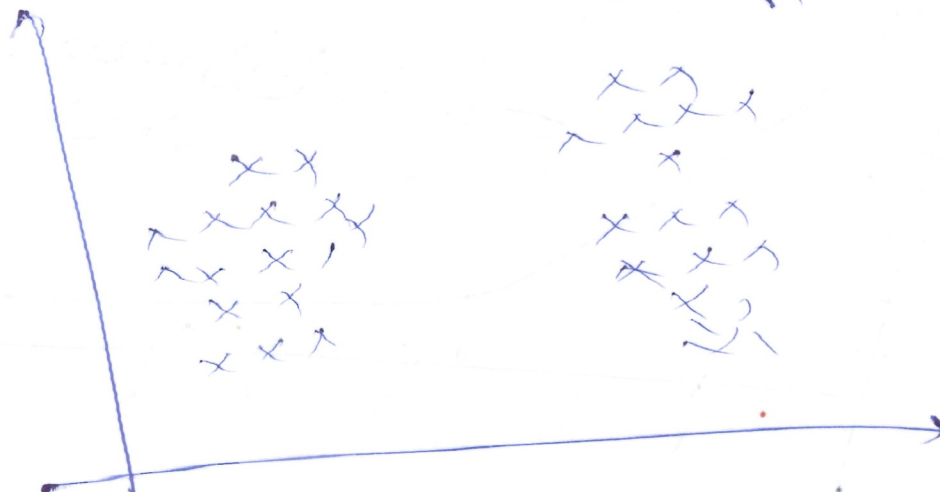
Euclidean distance

$$= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Manhattan distance

$$= |x_2 - x_1| + |y_2 - y_1|$$

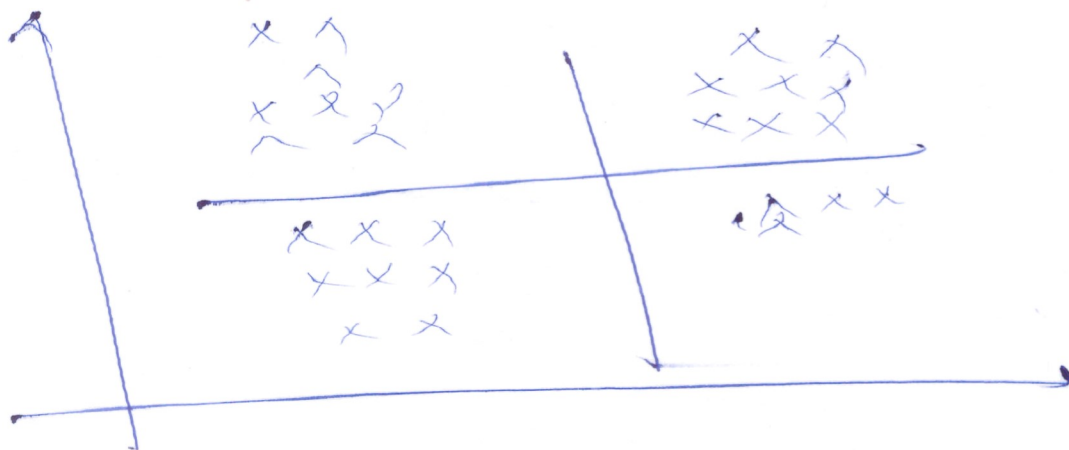
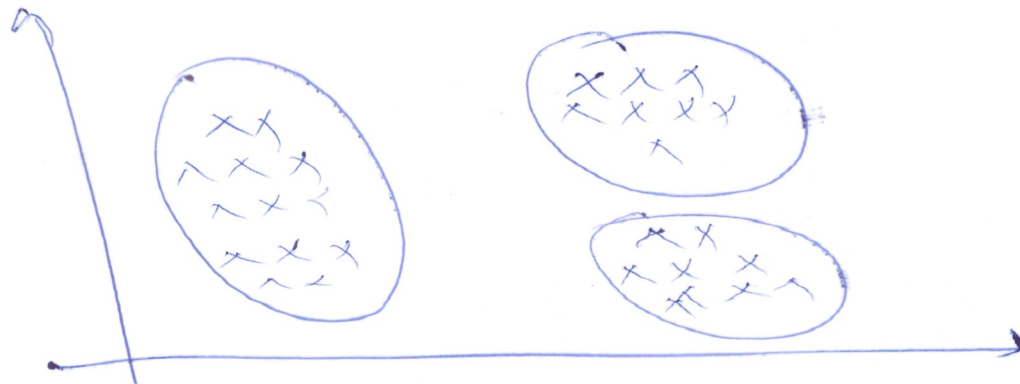
⇒ Random Initialization



P. 11

By applying K-means

②



K-means + Initialization